



TTK4147 Real-time Systems

01 Introduction

2026-08-17

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Who we are

- **Geir Mathisen**

Professor, Embedded Systems, at Department of Engineering Cybernetics

Previously acted as Senior Scientist at SINTEF Digital, been working with RT embedded systems within different application areas, among others 15 years within Smart Grid



- **Joakim Borge Hunskaar**

Assistant lecturer
Responsible for the exercises / lab



- **Mathias Harketstad Bæverfjord and
- Sverre Stalsberg**

Lab assistants

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Course information 2026

Canvas:

- Canvas will be used for general communication.
All information published on Canvas are expected to be known by the students.
- It will be a page on Canvas for questions and answers concerning the course and its content.

The course consists of lectures and exercises (lab)

- It is no one-to-one connections between the content of the exercises and the lectures.
- The exercises are partly meant to demonstrate lectured knowledge and partly to demonstrate other real time issues.

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Course information 2026 continue

Lectures:

- Physical lectures on Mondays 9:15 – 12:00 (often finished earlier) in H1
- The physical lectures will not be streamed or taped.
- Lecture slides from this year will be published ongoing on Canvas
Lecture slides from 2025 will also be published on Canvas.
Be aware that the administrative and organizing information given in 2025 slides will differ from what applies for 2026.

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Course information 2026 continue

Exercises:

- Physical lab exercises starting at week 35.
The lab shall be carried out in groups of two, see Canvas for information concerning forming the groups. You need to sign up for a group during week 34
- It will be a theory exercise in week 42 ("Semester assignment", Wednesday October 14th). This exercise shall be carried out one by one.
- To be able to take the exam, it is mandatory to accomplish at least 80 % of the lab exercises.
- The lab requires physical attendance



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Course information 2026 continue

Exam and course evaluation:

- The exam is planned to be November 21, 9:00 – 13:00
- Both the theory exercise and the exam must be passed to pass the complete course.
Both will be graded separately.

Course evaluation:

- Theory exercise 30 %
- Written (digital) examination (4 hours) 70 %
- Examination aids: D - No printed or hand-written support material is allowed. A specific basic calculator is allowed.

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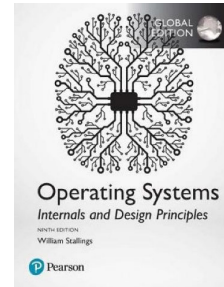
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Lecture Information

- Lectures are taken from the textbooks
 - Main book:
William Stallings: Operating Systems: Internals and Design Principles.
 - Secondary book:
Burns and Wellings: Real-time systems and their programming languages, 4th ed
This is not available in bookstores, but the relevant chapters are provided as PDFs on Canvas
 - MSIT 116D Distributed Systems, Chap 1-3, used in the lecture “Distributed Systems – Architecture” provided on Canvas
- See “Reading Material by Lectures” with different links to relevant URLs on Canvas



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Exercise information

- Compulsory assignments
>= 80% (>= 7 exercises) have to be done
- Exercise 0 – 8 are worked out in groups 2 by 2
Form groups on Canvas, choose wanted weekday.
- The Exercise texts 0 – 8 will be on Canvas
- The exercises will be held on **Thursday / Friday**
- The Theory Exercise, week 42, is to be worked out one by one.
Preliminary date and place: **Wednesday 2026-10-14, 18:00 – 20:15, Sentralbygg 2 S6**
- The Theory Exercise will be evaluated with a top score of 30 % points

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Lecture and exercise plan 2026

Date	Week	Lectures Monday 9:15 - 12:00 in H1- Main Building
2026-08-17	34	01 Introduction
2026-08-24	35	02 Hardware and operation systems
2026-08-31	36	03 Multiprogramming and Memory Management
2026-09-07	37	04 Scheduling
2026-09-14	38	05 Deadlocks, mixed criticality and freertos
2026-09-21	39	06 Priority Inversion
2026-09-28	40	07 Linux
2026-10-05	41	08 More Scheduling
2026-10-12	42	Question&Answer, Repetition
2026-10-19	43	09 QNX
2026-10-26	44	Guest lecture
2026-11-02	45	10 Dist. System Architectures
2026-11-09	46	11 Analysis of Physical systems
2026-11-16	47	Question&Answer, Repetition
2026-11-21	47	Exam 09:00 - 13:00

Date	Week	Exercises Thursday 8:00 - 12:00, Friday 8:00 - 12:00, Sanntidslabben, G203/G204
	34	
2026-08-20 / 2026-08-21	35	0- Basics and setup
2026-09-03 / 2026-09-04	36	1 - Time
2026-09-10 / 2026-09-11	37	2 - Memory
2026-09-17 / 2026-09-18	38	3 - Scheduling
2026-09-24 / 2026-09-25	39	4 - Concurrency
2026-10-01 / 2026-10-02	40	5 - AVR UC3
2026-10-08 / 2026-10-09	41	6 - AVR FreeRTOS
2026-10-14	42	Theory Exercise, In Sentralbygg 2 S6 at 18:00 (Starts 18:15)
2026-10-15 / 2026-10-16	42	7 - BRTT and Rpi
2026-10-22 / 2026-10-23	43	8 - BRTT and PREEMPT RT
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	45	

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Reference group students

We need (at least) 2 students to be in the student's reference group

If you will contribute to the reference group, please post me
Geir.Mathisen@ntnu.no some lines.

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Reflections

Talk with the person(s) sitting next to you (form groups of 2 or 3 persons) about:

- In your opinion, what is a real-time task (mark with '(T)') or a real-time system (mark with '(S)'), (examples from daily life)?
- What are your expectations for this course?

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Today

- Course content and Learning outcome
- Short introduction to Computer Components
- Short introduction to Real-time

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Course content

The course will deal with:

- Necessary computer hardware knowledge.
- Important issues from operating systems.
- Analysis of time requirements, resource allocation, principles of resource management within time bounds.
- Analysis and evaluation of requirements for runtime systems for embedded real-time systems.
- Distributed real-time and data acquisition systems.
- Analysis and custom for real-time applications.
- Practical skill in programming of real-time systems.

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Learning outcome

Knowledge:

- Extensive knowledge about principles and methods for resource-analysis for embedded- and real-time systems.
- Good knowledge of the relevant mechanisms and methods in operating systems and hardware that have influence on real-time aspects.
- Good knowledge about principles and methods for design and construction of embedded- and real-time systems.
- Knowledge about analysis of physical systems as a basis for design of real-time systems.

Skills:

- Practical skill in design, construction and implementation of embedded- and real-time systems

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What is an embedded system?

An embedded system is a computer system built into a larger device to perform a specific function or set of functions. An embedded system is designed for a particular task and is often optimized for size, cost, power consumption, and reliability.

Examples of Embedded Systems

- Microwave oven
- Car electronics: engine control units (ECUs), ABS brakes, airbags.
- Washing machines: manage water levels, temperature, and wash cycles.
- Routers: handle network traffic.
- Smart TVs: run user interfaces and streaming applications.
- Medical devices: such as pacemakers and infusion pumps.

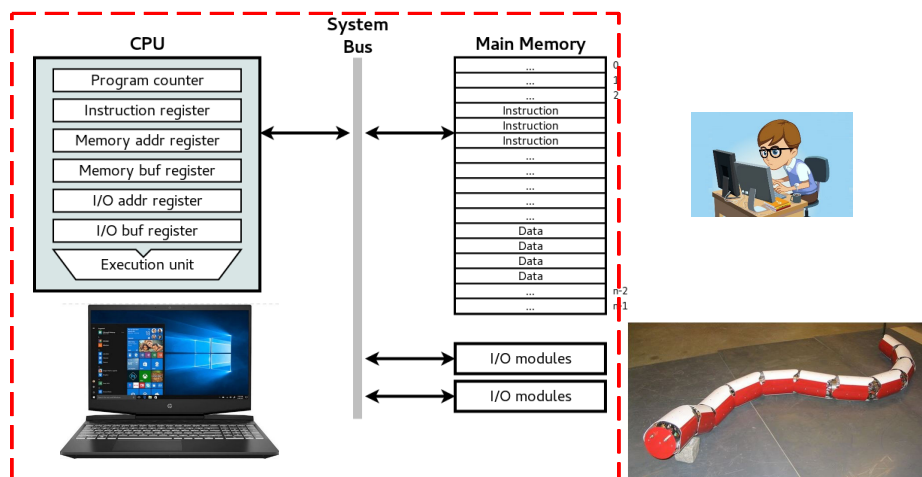
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Computer Components



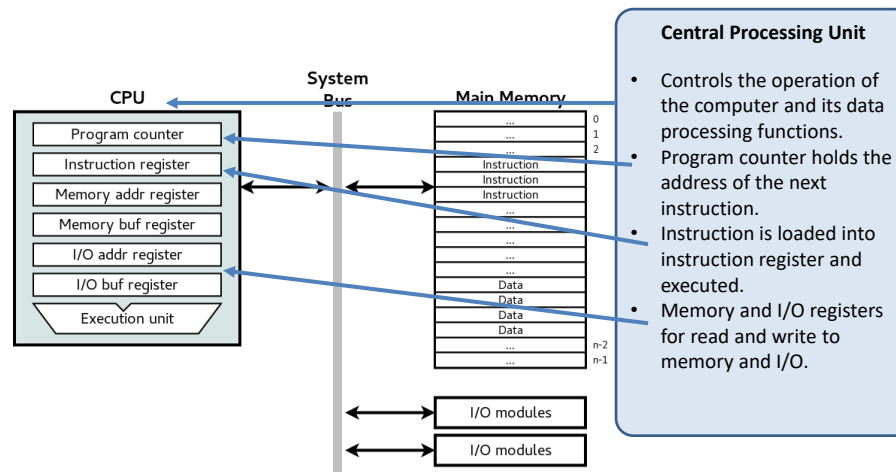
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Computer Components - CPU



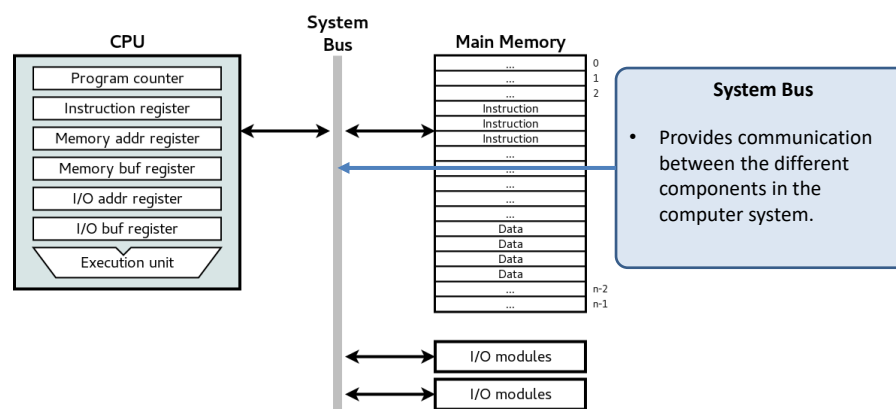
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Computer Components - System Bus



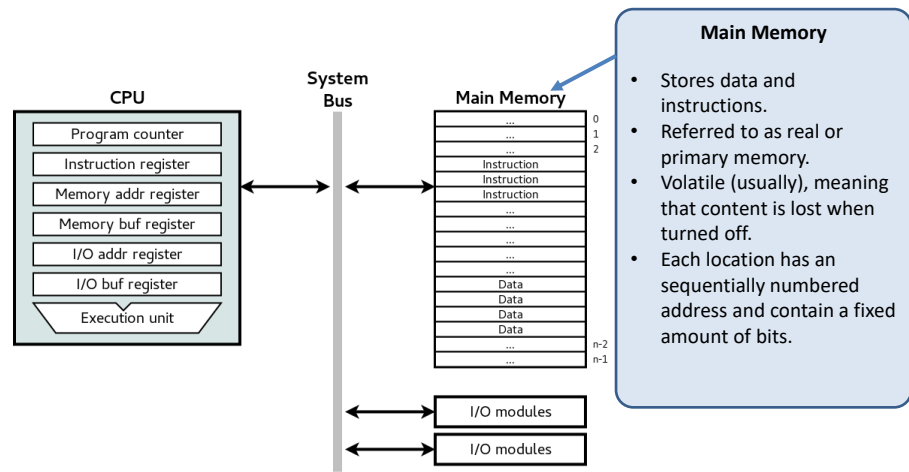
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Computer Components - Main Memory



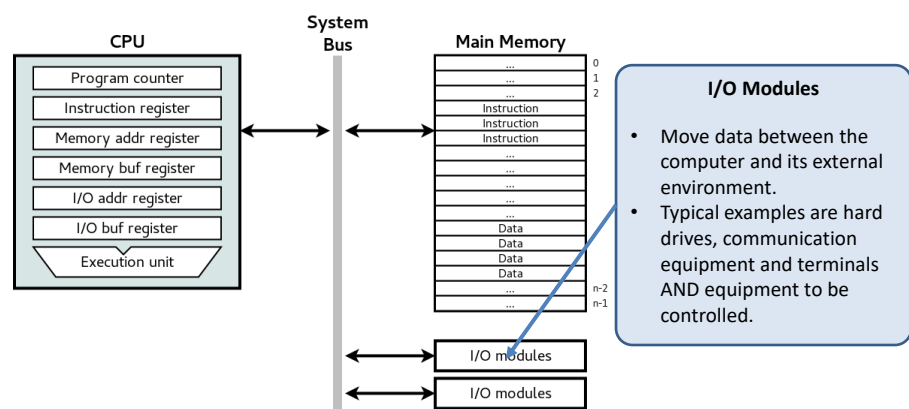
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Computer Components - I/O modules



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Evolution of the Microprocessors

The hardware revolution that has made desktop and handheld computing possible.

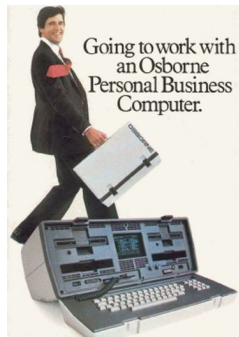
A central processing unit (CPU) on a single chip, external chips with memory and peripherals.

- The processor in "normal" PCs, servers etc.



Commodore 64, Jan 1982

Osborne 1,
April, 1981



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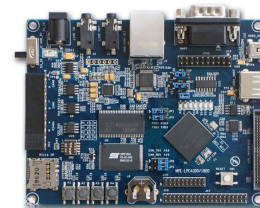
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Evolution of the Microprocessors cont.

A microcontroller unit (MCU) is a single chip with a processor, volatile and non-volatile memory and often some peripherals.

- Examples are Atmel (used by Arduino) and Freescale Semiconductor (Motorola).
- Often used in embedded systems.
- Vary from simple 8-bit microcontrollers to more complex 64-bit microcontrollers.
- Relatively easy to use and program.

Modern microprocessors are multiprocessors, with more than one processor (core) on each chip (socket).



Board with
microcontroller

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Microcontroller VS System on a Chip (SoC)

The difference between microcontroller and System on a Chip (SoC) is somewhat unclear.

- SoC are typically used for more complex microcontrollers that also include graphic processing units, digital signal processors etc.
- SoC is typically the brain in mobile phones, tablets, Raspberry Pi etc.



Raspberry Pi with a SoC

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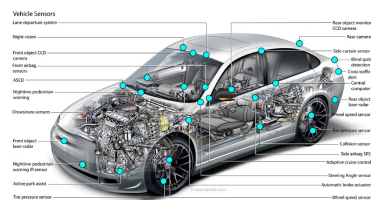
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The importance of microcontrollers in modern life

- About 99 % of all microcontrollers are used in embedded systems, most of them in RT systems.  What are the rest used for?
- The MCUs are embedded everywhere: cellular telephone, microwave oven, washer, dryer, dishwasher, coffee maker, refrigerator, TVs, video-game console, radios, remote control for the TV, cars



Can we do without the micro-chips?

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2020–2023 global chip shortage, History

(Which illustrates society's dependence on microprocessors)

- The demand for integrated circuits was greater than the supply
- Affecting > 169 industries
 - A major shortages and queues amongst consumers for video cards, video game consoles, cars and other electrical devices.
 - The global semiconductor shortages forced factory closures in a range of sectors, from cars to healthcare devices. This made more evident the extreme global dependency of the semiconductor value chain on a very limited number of actors in a complex geopolitical context.
- The shortage last to some degree into 2024

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Reasons for the 2020 – 2023 global chip shortage

(from Wikipedia)

- Main reason: The COVID-19 pandemic (global lockdowns) 
- China–United States trade war.
US placed restrictions on Semiconductor Manufacturing International Corporation (SMIC), China's biggest chip manufacturer. Forced companies to use other manufacturers like Taiwan Semiconductor Manufacturing Company Limited (TSMC) and Samsung. However, these companies were already producing at maximum capacity.  
- Taiwan drought
In 2021, Taiwan has its worst drought for more than 50 years. Chip manufacturers use large amounts of ultra-pure water to clean their factories and wafers. (TSMC used more than 63,000 tons of water a day) 

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Is there still a global chip shortage in 2026?

Yes, but the shortage has evolved.

What's still in shortage?

- DRAM (especially DDR4/DDR5)
- NAND flash
- High Bandwidth Memory (HBM) for AI workloads
- Some automotive components



What's causing it now?

- AI infrastructure boom
- Manufacturers shifting capacity to high-margin AI memory
- Geopolitical disruptions
- Limited factory expansion ("nobody's scaling up")



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Focus on chip dependency

- The global chip shortage following the Covid-19 pandemic has highlighted society's dependence on modern chips.
- Trade wars and geopolitical uncertainty have undermined confidence in a global market.
- Both the US and Europe are seeking to become independent of chip manufacturers on other continents and are building up their own industry.
- The EU wants to increase its share of the global microchip market from 10% in 2020 to 20% in 2030.
- More than 43 billion euros will be invested by 2030 to strengthen the European chip industry (The European Chips Act).



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Definition of Real-time Systems

"A real-time system is any information processing system which has to respond to externally generated input stimuli within a finite and specified period."

Young (1982)

"A real-time system is a system that is required to react to stimuli from the environment (including passage of physical time) within time intervals dictated by the **environment**."

Randell et al. (1995)

"In computer science, real-time computing (RTC), or reactive computing describes hardware and software systems subject to a "real-time constraint", for example from event to system response. Real-time programs must guarantee response within specified time constraints, often referred to as deadlines."

Wikipedia

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Real-time Systems in interaction with the environment

A real-time system should synchronize the RT-system itself internally and with the real world

- Different parts of the program (multi-tasking) should be in correct sequence.
- The program (system) should react on the environment rapid enough (before next event, before the user become impatient, ...).

If time problems:

One solution is to use a more powerful computer, another alternative is to make the program smarter.

- The system (cyber-physical system) must be realistic designed with respect to physical laws. The sum of the program and the physical system should be able to fulfil the requirements.

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Real-time Computing

Real-time computing is:

- Guaranteed or predictable response times.
- Worst-case execution time is more interesting than average execution time or throughput.
- Not really possible with general-purpose operating systems.

Real-time computing is NOT:

- A fancy way of saying “fast computing”.
- Irrelevant if hardware is fast enough.

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Philosophically, what is “Real-Time” (RT)?

Some thoughts

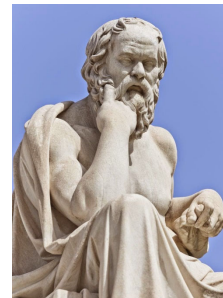
Is «RT» when something happens OR when it is sensed / interpreted / understood?

Is “RT” simply momentary NOW?

Has “NOW” passed? Is NOW NOW?

To sense, interpret and understand an event take some (various amount of) time. Delays are introduced.

Maybe we should talk about “Near Real-Time” (NRT)?



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“Real-Time” versus “Near Real-Time”

From Wikipedia:

The term "near real-time" or "nearly real-time" (NRT), in telecommunications and computing, refers to the time delay introduced, by automated data processing or network transmission, between the occurrence of an event and the use of the processed data.

...

The distinction between "near real-time" and "real-time" varies, and the delay is dependent on the type and speed of the transmission. The delay in near real-time is typically of the order of several seconds to several minutes.

Normally, we talk about “RT” when the delay has no significance for to interpret the results of a computation as happen “momentarily” (although we often should use the term NRT).

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Related Concepts

Embedded Systems

- A computer system with a dedicated function.
- Vary in scope from simple microcontrollers performing a single simple task to large distributed control systems.
- Often without “typical” screen and keyboard interface.
- Will often have real-time constraints, but not necessarily.

Cyber-Physical Systems

- An embedded system that (physically) interact with the real (physical) world.
- Interaction with the real world will typically mean real-time constraints.

Internet of Things (IoT), Industrial Internet of Things (IIoT)

- Networked embedded system.
- Often associated with small, low-powered devices, but may also refers to larger units (e.g. network connected working cells)

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Real-time Constraints

In a real-time system, one or more tasks will have real-time constraints. This means that they have to complete within a defined period or point in time.

Hard Real-time

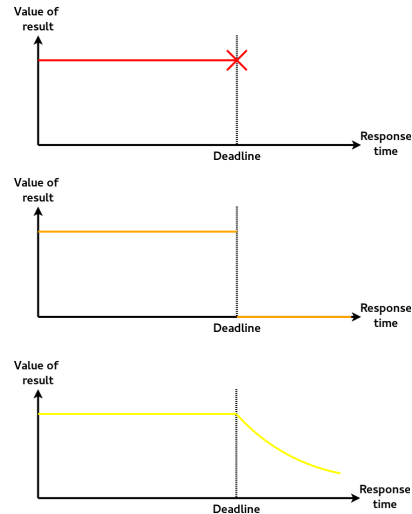
- Failure to complete within the deadline cause total system failure.
- Often associated with “catastrophic failure” (loss of life or large damage).

Firm Real-time

- Failure to complete within the deadline means that the result has no value.
- Repeated missed deadlines will lower the quality of the service provided by the system

Soft Real-time

- Failure to complete within the deadline means that the result has less value.



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Task Triggers

When is a task started or released?

Periodic:

- Task is performed at a regular time interval.

Aperiodic:

- Task is performed when an event occur.
- No limitations for when the event can occur.

Sporadic:

- Task is performed when an event occur.
- There are limitations on how often the event can occur.

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General Purpose VS Real-time

General purpose systems

- "Optimized" for speed and throughput.
- Usually comes with a wide range of applications, tools etc.
- Mouse, keyboard and screen.

Real-time systems

- "Optimized" for predictability.
- Usually a minimized system without anything that is not necessary
- Serial interface or SSH (specialized IO).

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Real-time Operating Systems (RTOS), to think about

1. Why are operating systems important for real-time?
2. Why do we need an operating system at all? (Or do we really?)
3. What makes a good RTOS?
4. Which operating systems are suitable for real-time and why?
5. When can we use a general purpose operating system? (why/why not?)
6. How to develop for embedded and real-time systems?

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Real-time Operating Systems

which we will touch into later

FreeRTOS

- Minimalistic operating system for microcontrollers that often do not use operating systems.
- Runs on many platforms.

QNX

- Proprietary and commercial operating systems for embedded and real-time systems.
- Stable and proven platforms.
- Expensive licenses.

Real-time variants of Linux

- Various methods for improving the real-time capabilities of Linux.
- RT-Linux, RTAI, Xenomai, preempt-RT.

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Once again

Reference group students

We need (at least) 2 students to be in the student's reference group

If you will contribute to the reference group, please post me
(Geir.Mathisen@ntnu.no) some lines.

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Old exam questions

[2021-10]

What are real-time systems optimized for? Select one alternative:

- Predictability ✓
- Throughput
- Accuracy
- Speed

[2020-1]

With your own words, describe a real-time system in NO MORE than two sentences.

Something along the lines of:

"A real-time system is any information processing system which has to respond to externally generated input stimuli within a finite and specified period."

"A real-time system is a system that is required to react to stimuli from the environment (including passage of physical time) within time intervals dictated by the environment."

"In computer science, real-time computing (RTC), or reactive computing describes hardware and software systems subject to a "real-time constraint", for example from event to system response. Real-time programs must guarantee response within specified time constraints, often referred to as deadlines."

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